

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 177 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	SN1 and SN2 Nucleophilic Substitution	91 PYQs	Reaction Mechanisms
Very High	Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	89 PYQs	Elimination & Aromatic

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SN1 and SN2 Nucleophilic Substitution	$\text{SN1: Rate} \propto [\text{RX}]$; $(3^\circ > 2^\circ > 1^\circ)$, Racemization , $\text{SN2: Rate} \propto [\text{RX}][\text{Nu}]$; $(1^\circ > 2^\circ > 3^\circ)$, Inversion
Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	Saytzeff Rule: Highly substituted, more stable alkene is the major product in dehydrohalogenation

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SN1 and SN2 Nucleophilic Substitution (Very High Yield — 91 PYQs)

Core Principle: Kinetics, stereochemical outcomes, carbocation rearrangements in SN1, and concerted backside displacement in SN2.

■ **Caution / JEE Pitfall:** Polar PROTIC solvents (e.g. H_2O , ROH) stabilize carbocations and favor SN1. Polar APROTIC solvents (e.g. acetone, DMSO) favor SN2 by desolvating nucleophiles.

■ **Speed Trick:** Allylic and benzylic halides undergo BOTH SN1 (resonance-stabilized carbocation) and SN2 (low steric hindrance) at remarkably high rates.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement (I) : Benzyl chloride reacts faster in $\text{S}_{\text{N}}1$ mechanism than ethyl chloride. Statement (II) : E..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Given below are two statements : Statement I : 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): " $\text{Ph-CH=CH}_2 \xrightarrow{\text{HBr}} \text{PhCH}_2\text{CH}_2\text{Br}$ Product Consider the above reaction A. The reaction proceeds through a more stable..." [Ans: N/A]

3.2 Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions (Very High Yield — 89 PYQs)

Core Principle: β -elimination mechanism, Saytzeff vs Hofmann regioselectivity, and unreactivity of haloarenes toward nucleophilic substitution.

■ **Caution / JEE Pitfall:** Haloarenes (like chlorobenzene) are extremely UNREACTIVE toward nucleophilic substitution due to partial double bond character (C-Cl resonance) and sp^2 carbon.

■ **Speed Trick:** Presence of electron-withdrawing groups (like $-NO_2$) at ortho- and para-positions dramatically increases reactivity of haloarenes toward nucleophilic substitution.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : Due to increase in van der Waals forces, the order of boiling points is $CH_3CH_2CH_3 > CH_3CH_2CH_2CH_3 > CH_3CH_2CH_2CH_2CH_3$..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Consider the following compounds Arrange these compounds in the increasing order of reactivity with nitrating mixture...." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is : $CH_3Br + Nu^- \rightarrow$..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is : $CH_3Br + Nu^- \rightarrow CH_3Nu + Br^-$

- (A) $ClO_4^- > CH_3COO^- > OH^- > PO_4^{3-}$
- (B) $OH^- > PO_4^{3-} > CH_3COO^- > ClO_4^-$ **[Correct]**
- (C) $CH_3COO^- > PO_4^{3-} > OH^- > ClO_4^-$
- (D) $PO_4^{3-} > OH^- > CH_3COO^- > ClO_4^-$

Correct Answer: B

Solution Analysis: For CH_3Br (a methyl halide), the reaction proceeds by S_N2 , so the rate increases with nucleophilicity of Nu^- . For oxygen nucleophiles in the same row, nucleophilicity roughly follows basicity, and any resonance stabilization of the anion reduces nucleophilicity: OH^- : strong base, charge localized $\Rightarrow$ best nucleophile
 PO_4^{3-} : resonance-stabilized $\Rightarrow$ less nucleophilic than OH^-
 CH_3COO^- : even more resonance delocalization over two oxygens $\Rightarrow$ weaker nucleophile
 ClO_4^- : highly resonance-stabilized, essentially non-nucl

Benchmark Problem 2: JEE Main 2024 (Online) 27th January Morning Shift (Q#4)

The correct statement regarding nucleophilic substitution reaction in a chiral alkyl halide is ;

- (A) Retention occurs in S_N1 reaction and inversion occurs in S_N2 reaction.
- (B) Racemisation occurs in S_N1 reaction and retention occurs in S_N2 reaction.
- (C) Racemisation occurs in both S_N1 and S_N2 reactions.
- (D) Racemisation occurs in S_N1 reaction and inversion occurs in S_N2 reaction. **[Correct]**

Correct Answer: D

Solution Analysis: The correct statement regarding nucleophilic substitution reaction in a chiral alkyl halide is Option D. Let's break down why this is the case by understanding both the S_N1 and S_N2 reactions and their effects on the chirality of the alkyl halide. S_N2 Reaction: The S_N2 (Substitution Nucleophilic Bimolecular) reaction is a one-step process where the bond formation (nucleophile attacking the substrate) and the bond-breaking (leaving group leaving the substrate) processes occur simultaneously. This reaction type is associated with an inversion of configuration at the chiral center.